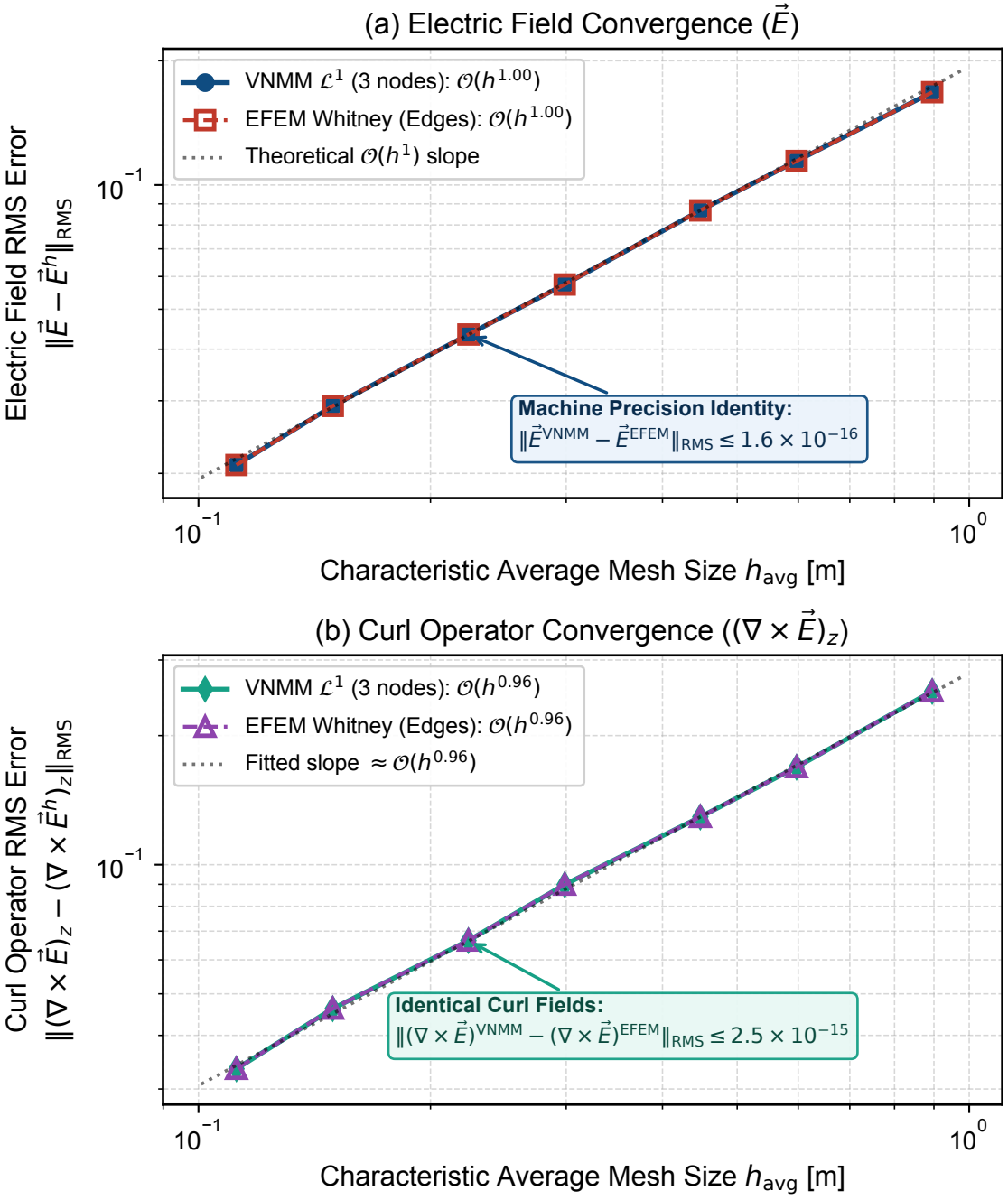
(a) Electric Field Convergence ($\vec{E}$)
VNMM $\mathcal{L}^1$ (3 nodes): $\mathcal{O}(h^{1.00})$
EFEM Whitney (Edges): $\mathcal{O}(h^{1.00})$
Theoretical $\mathcal{O}(h^1)$ slope
Machine Precision Identity:
$\|\vec{E}^{\text{VNMM}} - \vec{E}^{\text{EFEM}}\|_{\text{RMS}} \leq 1.6 \times 10^{-16}$
Electric Field RMS Error
$\|\vec{E} - \vec{E}^h\|_{\text{RMS}}$
10^{-1}
10^{-1}
10^{0}
Characteristic Average Mesh Size h_{avg} [m]
(b) Curl Operator Convergence ($(\nabla \times \vec{E})_z$)
VNMM $\mathcal{L}^1$ (3 nodes): $\mathcal{O}(h^{0.96})$
EFEM Whitney (Edges): $\mathcal{O}(h^{0.96})$
Fitted slope $\approx \mathcal{O}(h^{0.96})$
Identical Curl Fields:
$\|(\nabla \times \vec{E})^{\text{VNMM}} - (\nabla \times \vec{E})^{\text{EFEM}}\|_{\text{RMS}} \leq 2.5 \times 10^{-15}$
Curl Operator RMS Error
$\|(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z\|_{\text{RMS}}$
10^{-1}
10^{-1}
10^{0}
Characteristic Average Mesh Size h_{avg} [m]